

Hynes Birmingham

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EDUCATION

University of Connecticut

B.S. Biomedical Engineering | Minor in Computer Science | Concentration: Systems & Instrumentation

GPA: 3.41 | CAPS Research Scholar

Storrs, CT

May 2029

RESEARCH & INSTRUMENTATION EXPERIENCE

Ryberg Lab — University of Connecticut, Allied Health Sciences

Undergraduate Researcher — CAPS Research Scholar

Storrs, CT

Nov 2025 – Present

- Designed and built a full-stack automated bacterial colony counting instrument for a university microbiology lab: Python/OpenCV computer vision pipeline (96.3% count accuracy, F1 0.957, <0.10% false-positive rate on 820-plate synthetic validation set) deployed as a Flask web application on embedded Linux (Raspberry Pi 5)
- Designed a custom power-control PCB in KiCad for LED backlight management (board ordered); end-to-end instrument development spanning hardware design, embedded Linux deployment, and computer vision — analogous to design verification workflows in regulated medical device development
- Engineered a continuous self-validating testing pipeline with synthetic data generation, automated execution, and full reproducibility tracking (git commit hash, environment snapshot, input hash); 59/59 unit tests passing across 6,345 SLOC
- Built a two-layer anomaly detection system (Z-score/IQR statistical baseline + Random Forest classifier) on a structured 39-organism × 11-media-type profile system with per-run provenance tracking; accepted to present at UMBC Undergraduate Research Conference (Sep 2026)

TECHNICAL PROJECTS

Pharmacokinetic Simulator

Nov 2025 – Dec 2025

- Built a modular two-compartment ODE model in Python supporting IV bolus, IV infusion, and oral dosing routes; RK45 solver validated against closed-form analytic solution to <0.001% error on AUC and $t_{1/2}$; single-dose simulation runtime 3.03 ± 0.03 ms

FocusGuard — Multi-Modal Sensor Fusion System

Jan 2026 – Present

- Multi-threaded producer-consumer system fusing real-time computer vision (MediaPipe face mesh, blink rate) with keyboard behavioral signals; 26 ms median sensor-to-score latency at 30 fps, zero crashes over 6-minute stability soak, 10/10 unit tests passing

STM32 Bare-Metal GPIO Driver

Jan 2026 – Present

- Register-level GPIO driver for STM32F411RE in C; 9 MMIO registers configured across 3 peripherals, no HAL abstraction, 752 B flash image, logic analyzer verification

HuskyBook — Peer-to-Peer Campus Directory

Jan 2026 – Present

- Node.js + Express REST API (16 endpoints, 5-table SQLite schema); 32/32 tests passing, 0.07–0.16 ms read latency, 1,724 SLOC

LEADERSHIP & COMMUNITY

BRIDGE STEM Coach — Vergnano Institute of Impact, UConn

Jun 2026 – Aug 2026

- Coached 43 incoming first-generation and underrepresented STEM students through UConn's BRIDGE transition program
- Independently designed and delivered a biotechnology and instrumentation module to a cohort of 8 students covering bacterial transformation, fluorescence imaging, and quantitative data analysis
- Built a companion Python tool in Google Colab enabling students without local software access to fully process and visualize their own experimental data

Learning Community Tutor — Scholars House, UConn

Fall 2025 – Spring 2026

- Provided STEM tutoring to residents of Scholars House, a living-learning community for Black men in higher education; 2–3 sessions/week over 12 weeks, contributing to the community's 3.0 GPA goal by Spring 2026

Innovation Zone — University of Connecticut

Students as Makers (SAMs) — Founding Cohort

Storrs, CT

Feb 2025 – Dec 2025

- Member of the founding cohort of UConn's Students as Makers program; fabricated 7 projects for improvement of the Innovation Zone and related services utilizing 3D printing, manufacturing, and power tools, saving over \$2,300 compared to purchasing equivalent equipment or products

SKILLS

Biomedical & Instrumentation: Computer vision for biological measurement, pharmacokinetic modeling, embedded Linux instrument deployment, KiCad PCB design, sensor fusion, physiological signal processing

Programming: Python, C, MATLAB, OpenCV, scikit-learn, SciPy, pandas, NumPy, Flask/REST APIs, multi-threaded systems, Node.js, Git

Data & Validation: ETL pipelines, automated testing, provenance tracking, structured data logging, OOP design, version control